TOP 125 LATEST DATA ENGINEER INTERVIEW QUESTIONS OF 2026

SQL & Pyspark Questions

1. Return matching records from 2 tables which also contain Null values.
2. Return all records from Left Table which matches with Right Table.
3. From a Travel history table containing User Id, City and Travel Date, identify the source and destination city of each journey?
4. Find 4th highest salary for each department.
5. What is the difference between Rank & Dense_Rank?
6. From a large data set having duplicate records, keep only latest records?
7. Count Null values first column wise and then row wise?
8. How do you handle Json Data?
9. How do you handle a slow spark job?
10. How do you optimize a slow SQL query?
11. Why can a job still be slow even after filtering 80% of data?
12. How does Spark optimize filters on Delta tables?
13. What happens internally during repartition?
14. How would you handle a skew during production?
15. How do you remove Duplicates?
16. What if Duplicates have different timestamps?
17. Find Minimum, Maximum and Average Salary for every department in a company?
18. Find moving average sales for 3 Days (Friday, Saturday and Sunday) in a quarter?
19. From a flight database, find out total delay for each flight in a week?
20. How do you calculate transaction volume for a merchant?
21. Calculate Daily Cumulative Balance for merchant?
22. Fetch out Top 3 geographies with highest customer base?
23. How do you handle Fault Tolerance using Spark?
24. What is the difference between Job, Stage and Task? How Spark decides number of tasks per stage?
25. What is difference between Broadcast Join vs Sort Merge Join?
26. Explain all types of Join with a simple use case?
27. How do you execute a PySpark code on a cluster?
28. How do you decide cluster configuration for your PySpark job?
29. How do you Debug a failed Spark job?
30. Convert rows into column using SQL or Pyspark?
31. Explain narrow vs wide transformations?
32. Convert SQL query into PySpark DataFrame code and find out the highest salary per department in the employee table.
33. Find duplicate records in a DataFrame and remove them.
34. How do you apply Join on 3 columns.
35. Explain SCD Type 2 with a use-case.
36. How do you count number of Credit vs Debit Transactions per user. What insight can you get from this?

37. Identify and flag a user who undergoes 3 transactions within 60 seconds?
38. How do you optimize transformations in PySpark?
39. Difference between Repartition vs Coalesce.
40. What happens internally while running a PySpark job?
41. Ways to handle Data skew, OOM errors and Streaming failures.
42. Brief about Spark 3.0 features (Adaptive Query Execution and Auto skew handling).
43. Row vs Columnar formats (Parquet, ORC, Avro).
44. Best practices for Partitioning & bucketing
45. Explain Window/analytical functions
46. Explain UDF vs native functions (performance impact)
47. Explain Driver vs Executor roles?
48. Memory management and JVM GC behaviour.
49. Interpret GC logs and identify memory pressure issues
50. What is DAG creation and explain task execution.

Kafka Questions

1. Explain your real time experience in Kafka?
2. How do you tackle flaws?
3. How do you ensure high throughput and low latency in Kafka?
4. Explain designing of real time payment processing pipeline using Kafka?
5. How do you handle schema evolution from Kafka to Delta Lake.

Spark Structured Streaming Questions

1. Checkpointing (offsets, state, metadata)
2. Watermarking & late data handling
3. State management across micro-batches
4. How is Spark Structured Streaming different from DStreams.
5. Debug issues (OOM errors, Skewed partitions, Slow streaming jobs)

Azure Cloud / Databricks Questions

1. How do you pass parameter from one Notebook to another Notebook?
2. If records are moved from source to target table, and source is deleted. How do you identify the deleted records? (Upsert & Delete detection)
3. Explain Databricks Architecture?
4. A Delta table is grown to double over time. How would you optimise the performance of this Delta table?
5. In which all places have you used Databricks ecosystem during production?
6. What is the difference between Delta Lake vs Parquet?
7. Difference between Full load and Incremental load?

8. Design a pipeline which can handle incremental load using Databricks.
9. If a spark job is running slow on Databricks, how will you Optimize it?
10. What do you mean by small file problem and how do you solve this problem?
11. How partitions count is decided? What will you do if partitions are too low?
12. How do you process a data of 5TB size in Databricks?
13. How do you design a pipeline with the help of Databricks?
14. How do you integrate Databricks and Azure Data Factory?
15. Explain with example, ACID properties in Delta tables, Delta Lake?
16. How Databricks interact with Data Lake?
17. Which services will you use for ingestion, transformation and storage?
18. How will you design a Data Warehouse? Where will you use SCD Type 1 and SCD Type 2?
19. What is primary use case difference between ADF vs Databricks.
20. Describe Medallion Architecture (Bronze → Silver → Gold)
21. Clearly explain data flow and design decisions
22. Explain Cluster types (job vs all-purpose)
23. Explain Autoscaling & VM sizing
24. Explain Driver vs Worker nodes
25. Explain Managed vs External tables
26. Explain Init scripts
27. Explain delta_log metadata
28. What is Time travel & retention?
29. Optimistic concurrency control
30. Compare Parquet vs Avro vs Delta vs Iceberg

AWS Cloud Questions

1. Explain how will you build end-to-end data pipeline using AWS services?
2. Which services will you use for ingestion, transformation and storage?
3. How AWS Glue works, and explain a real case example from any of your previous projects?
4. When will you use Glue and when Lambda?
5. How do you orchestrate multiple ETL jobs in AWS?

Project Related Questions

1. Give a brief about your last project and what was your individual responsibilities?
2. Explain End-to-End architecture of your previous project?
3. What was the data volume you were handling? Did you face any major challenge handling it? How did you optimize it?
4. Which part of pipeline were you handling solely?
5. Can you redesign your project using latest Databricks tools?
6. Have you worked on any Azure cloud project in past? Explain your role?
7. What was the design layout of your tables?
8. How many task, stages and partitions were created in your last project?
9. Did you face any sudden job delay in your previous project. How did you debug it?

10. Which all Integrations have you done till now?
11. Which all optimization techniques have you used in your last project?
12. How did you load huge csv files in any of your previous projects?
13. Which type of errors you have come across in your last project?
14. If you identify 5X times increase in data volume, what will be your steps to scale?
15. Did you come across duplicate records in your previous project? How do you fix this issue?

Scenario-based/System Design Questions

1. Design a system which is capable of handling millions of transactions every second?
2. Design a fraud detection pipeline which functions in real time?
3. Did you use any optimization technique in your previous project? Give a brief on it?
4. What will be your action plan when your pipeline fails suddenly?
5. How did you deal with deployment. What major challenges did you face?

Managerial Questions

1. Did you come across any conflict situation with your team members earlier. How did it get resolved?
2. How do you distribute tasks and handle dispute within team members?
3. If you face a major misalignment with a big client related to project timelines. How do you address this?
4. If you were given a chance to handle a team, what would be your plan to ensure each member deliver their best potential?
5. If you are asked to handle a real time end to end pipeline, for which part will you need an external help?
6. Which role do you see yourself in, after 3 years, if you get selected?
7. How many members were there in your last project team? Explain your exact role within your team?
8. How did you maintain clarity for a client requirement, with your client and internal team?
9. Which tool or technology did you update yourself on recently, which was new to you?
10. How do you talk to a client who is facing challenges in cost optimization?